VLP/1: identity, authorization, and continuity in verifiable computational communication

VCore Link Protocol · identity, authorization, and continuity

Rafael Felippe Voigt
VoigtCore Tecnologia I.S., Brazil
ORCID: 0009-0001-1091-8398
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VOIGTCORE TECNOLOGIA I.S.

Abstract

This paper presents VLP/1 (VCore Link Protocol), a protocol that is proprietary in its semantics and communication contract and was designed to relate persistent computational agents without conflating network presence, identity, trust, authorization, and data access. VLP/1 does not create proprietary cryptography: it uses established mechanisms such as TLS 1.3, mutual authentication, and Ed25519 signatures as foundations for channel security and authenticity. Its contribution lies in organizing independent decisions, preserving information nature and temporal provenance, and applying data minimization by default. An experimental artifact was evaluated in two local processes using ephemeral certificates and synthetic data. The tests verified mutual authentication, signed envelopes, ACK/NACK, scope control, idempotency, and replay rejection without enabling remote discovery, real enrollment, or operational telemetry. This paper discloses the scientific model and aggregated results while withholding the details required to reproduce an operational deployment. The "VCore VLP/1" computer-program registration request was filed with Brazil's National Institute of Industrial Property (INPI) on August 26, 2026, under process 512026007083-0. Filing does not constitute a patent, technical endorsement, or conversion of VLP/1 into an open standard.

Keywords: VLP/1; VCore Link Protocol; computational identity; persistent agents; authorization; temporal provenance; Zero Trust; mTLS; Ed25519; replay protection; data minimization.

1. Introduction

Protecting a channel does not by itself resolve the meaning of communication. A session may be encrypted while leaving essential questions unanswered: which entity is speaking, on behalf of which installation, under which authorization, about what kind of data, and with what historical continuity? In persistent systems, addresses change, processes restart, keys rotate, and products may be reinstalled. Technical similarity is not sufficient proof of continuity.

VLP/1 was created to address this gap between transport security and organizational meaning. Its guiding question is:

> *How can verifiable communication be established among persistent computational agents without turning discovery into trust, authentication into authorization, or missing data into a false conclusion?*

The protocol belongs to the VCore ecosystem, where distinct products retain distinct responsibilities. Pulse observes operational condition; BlackBox preserves evidenced actions and trajectories; VCore Server composes identity, context, and organizational memory. VLP/1 is the structured communication layer among these components, not a replacement for them.

2. Scientific contribution and proprietary boundary

The VCore contribution is not the invention of TLS, certificates, Ed25519, or general Zero Trust concepts. These are external standards and bodies of knowledge used under their own specifications. VLP/1 contributes a semantic architecture that combines established mechanisms under four verifiable rules:

1. discovery does not grant trust;
2. verified identity does not grant authorization;

3. authorization does not grant unrestricted data access;
4. unavailable information does not become zero, observed absence, or fact.

VLP/1 is proprietary in its identity model, message contracts, association rules, states, data policies, and integration with the VCore ecosystem. Publication of this paper allows the proposal to be understood, analyzed, and cited. It does not grant an implementation license, publish the complete operational contract, or turn VLP/1 into an IETF, W3C, or ISO standard.

3. Public VLP/1 model

3.1 Separate entities

The protocol distinguishes entities that may be related but are not equivalent: protocol agent, cryptographic identity, device, product instance, organizational Avatar, and authorizing organization. A hostname, IP address, MAC address, user, or license does not automatically create a VLP identity. Continuity requires evidence and explicit association.

3.2 Decision gates

The public conceptual flow is deliberately simple:

```
OBSERVED PRESENCE
        ↓
VERIFIED IDENTITY
        ↓
APPROVED ENROLLMENT
        ↓
SCOPE AUTHORIZATION
        ↓
DATA POLICY
        ↓
ACCEPTED SESSION AND MESSAGE
```

Each transition depends on its own evidence. An agent may be prepared but disconnected; identified but unenrolled; authorized for heartbeat but prohibited from transmitting telemetry. This separation reduces implicit trust and makes acceptance or rejection explainable.

3.3 Nature and provenance

Communicated information preserves its nature: `OBSERVED`, `DECLARED`, `INFERRED`, or `UNAVAILABLE`. Nature does not replace availability, trust, or authorization. Where the applicable contract permits, a message retains references to origin, time, policy, and correlation. This brings communication closer to temporal memory without turning the protocol into a database.

3.4 Two security layers

The channel and the message serve complementary roles. TLS 1.3 with mutual authentication protects the session. The VLP signature protects authorship and logical integrity of the envelope. Nonce, sequence, message identifier, scope, and idempotency support replay and conflict detection. Exact operational parameters remain outside this publication.

3.5 Minimization by default

The experimental profile uses allowlists. Capabilities describes what an agent may offer; heartbeat describes agent continuity. Neither needs to carry file content, credentials, human messages, or detailed machine telemetry. Agent presence is not treated as proof of human activity or complete device health.

4. Controlled evaluation

The artifact was built and evaluated incrementally: first in one process; then with identity and envelope controls; next between two local processes using TLS 1.3 and mTLS; and finally through an observational bridge exposing only sanitized state to the runtime. Certificates were ephemeral, communication used loopback, and inputs were synthetic fixtures. No remote server, real discovery, operational enrollment, real Pulse data, or production certificate was used.

Verification set	Result	Meaning
Local mTLS transport	30/30	Local channel and negative cases passed
Observational bridge	30/30	Sanitized state without remote activation
Consolidated VLP suite	114/114	Identity, contracts, scopes, replay, and transport
Functional regression	132/132	No regression observed in the evaluated gate
Portable security	5/5	Checks without a real operational profile

The counts represent passed test cases, not a statistical sample, security certification, or proof that vulnerabilities are absent. They support local architectural feasibility.

5. Who may use it, and where?

VLP/1 was initially designed for authorized VCore components, including Pulse agents, BlackBox, and VCore Server. Its architecture is also relevant, as a research object or licensed integration, to:

- persistent agents on computers, edge devices, and on-premises environments;
- services that must separate technical identity from organizational identity;
- hybrid infrastructures requiring explicit enrollment and authorization;
- observability systems that must express unavailability without fabricating zero;
- organizational memory and digital twins that preserve provenance and continuity;
- security, distributed-systems, and agent-governance laboratories.

Anyone may study and cite this public description. Use of the name, proprietary contracts, or a compatible implementation depends on authorization or licensing from the rights holder. VLP/1 was not designed as a VPN, remote shell, unrestricted administration protocol, or mandatory telemetry bus.

6. Frequently asked questions

Is VLP/1 a new protocol?

Yes, as an application protocol and semantic architecture for the VCore ecosystem. It uses external cryptographic and transport standards while defining its own identity, authorization, provenance, and continuity contract.

Is VLP/1 an open standard?

No. This publication is a public scientific note, not a complete open specification. The protocol remains proprietary.

Does VLP/1 replace TLS?

No. TLS 1.3 protects the channel. VLP/1 operates above it to organize identity, authorization, state, and message meaning.

Does discovering an agent mean trusting it?

No. Observed presence, verified identity, enrollment, authorization, and data policy are different decisions.

Does heartbeat transmit telemetry?

Not necessarily. In the evaluated profile, heartbeat proved agent continuity while operational telemetry remained disabled.

Does the protocol automatically identify a person or machine?

No. Hostname, IP address, MAC address, user, or license is not automatically converted into a VLP identity, Device, or Avatar.

Can VLP/1 be used outside VCore?

The model may be studied in other domains. A compatible implementation or commercial use of proprietary contracts requires an agreement with the rights holder.

Is the INPI filing a patent?

No. It is a computer-program registration request. It is distinct from patentability, security certification, and technical standardization.

7. Limitations and responsible disclosure

The public validation considered here was performed on one computer, between local processes, and with ephemeral certificates. It did not evaluate cross-network communication, prolonged operation, production PKI, reconnection storms, operational key rotation, load at scale, or a complete independent security review.

To preserve security and intellectual property, this paper does not publish canonical serialization, signature bytes, identifier derivation, replay windows, complete enrollment formats, keystore policies, endpoints, ports, certificates, source code, or deployment configuration. These elements are not necessary to understand the scientific hypothesis and results presented here.

The "VCore VLP/1" computer-program registration request was filed with the Brazilian National Institute of Industrial Property on August 26, 2026, under process 512026007083-0. The reference identifies the software filing. It does not imply a granted patent, INPI endorsement of the scientific model, security certification, or public permission to implement the protocol.

8. Conclusion

VLP/1 proposes that secure communication among persistent agents requires more than encrypting a connection. Identities, decisions, information natures, and continuity must remain separate and auditable. The local evaluation showed that an agent can hold a verifiable identity, exchange minimized messages, receive ACK/NACK, and resist replay without transmitting real operational data.

The central contribution is a semantic discipline: presence is not trust; authentication is not authorization; authorization is not total access; and unavailability is not a negative fact. Under these rules, VLP/1 acts as a controlled bridge among identity, continuity, and evidence. Public disclosure enables the proposal to be understood and assessed while preserving the operational and proprietary elements required for secure implementation.

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Scope statement

This document is a public scientific-technical note. It contains no credentials, private keys, operational certificates, production endpoints, source code, or authorization to activate VLP/1. VLP, VCore, VCore Pulse, and VCore BlackBox are designations of the VoigtCore ecosystem. External standards remain subject to their own specifications and licenses.